Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe Henriques Dairy

Physical address of dairy:

8255 Maze BLVD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1909

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X012-X060-X002-XXXX

B. OPERATORS

Henriques, Joe

Operator name: Henriques, Joe

Telephone no.: (209) 527-6511 (Landline) (Cellular)

736 N Gates RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Henriques, Joe

Legal owner name: Henriques, Joe

Telephone no.: (209) 527-6511 (Landline) (Cellular)

736 N Gates RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	135	90	78	75
Number under roof	243	37	0	0	0	0
Maximum number	325	50	180	120	105	100
Average number	243	37	135	90	78	75
Avg live weight (lbs)	1,400	1,450	100	800		

Predominant milk cow breed: Holstein

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 9,026.07 *tons per reporting period*

Total nitrogen from manure: 115,183.42 *lbs per reporting period*

After ammonia losses (30% loss applied): 80,628.39 *lbs per reporting period*

Total phosphorus from manure: 18,443.34 *lbs per reporting period*

Total potassium from manure: 45,578.84 *lbs per reporting period*

Total salt from manure: 122,924.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,750,000 *gallons*

Total nitrogen generated: 32,520.47 *lbs*

Total phosphorus generated: 5,006.50 *lbs*

Total potassium generated: 70,690.08 *lbs*

Total salt generated: 615,084.92 *lbs*

	9,750,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	9,750,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 3	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/29/2023	Solid commercial fertilizer *UN32*	9.74 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
06/27/2023	Solid commercial fertilizer *4-0-0*	8.30 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	6,890.70	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	6,890.70	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	28	28	2	process wastewater	X012-X600-X002-XXXX
Field 2	55	55	2	process wastewater	X012-X600-X006-XXXX X012-X600-X007-XXXX X012-X600-X008-XXXX
Field 3	60	60	1	manure	X012-X600-X002-XXXX
Field 4	56	56	1	manure	X012-X600-X002-XXXX
Totals for areas that were used for application	199	199	6		
Totals for areas that were not used for application					
Land application area totals	199	199	6		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	379.00 *ton*	As-is		56.4	4,010.00	1,180.00	14,000.00		0.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.54	108.56	31.94	379.00	23.61

Field 1

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	664.00 *ton*	As-is		69.8	3,500.00	450.00	2,940.00		2.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.71	166.00	21.34	139.44	300.79

Field 2

Field name: Field 2

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	691.00 *ton*	As-is		62.2	6,090.00	1,100.00	14,400.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.56	153.03	27.64	361.83	864.33

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	1,346.00 *ton*	As-is		70.0	3,900.00	4,400.00	2,700.00		1.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.47	190.89	215.36	132.15	278.99

Field 3

Field name: Field 3

12/28/2016: Almond, in shell

Crop: Almond, in shell Acres planted: 60 Plant date: 12/28/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	341.00 *ton*	As-is		8.6	20,100.00	1,320.00	25,000.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	5.68	228.47	15.00	284.17	945.41

Field 4

Field name: Field 4

12/19/2017: Almond, in shell

Crop: Almond, in shell Acres planted: 56 Plant date: 12/19/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	284.00 *ton*	As-is		8.2	21,500.00	1,230.00	21,300.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	5.07	218.07	12.48	216.04	828.69

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/29/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.69	16.74	243.64	1,122.10	750,000.00 *gal*
Application event totals		80.69	16.74	243.64	1,122.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	53.59	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.59	11.67	179.04	1,739.04	750,000.00 *gal*
Application event totals		62.59	11.67	179.04	1,739.04	

Field 1 - 06/27/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.85	12.83	196.95	1,912.94	825,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	58.95	5,346,000.00 *gal*
Application event totals		68.85	12.83	196.95	1,971.89	

Field 1 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	48.23	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	144.85	12.69	141.43	2,152.56	675,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	48.23	4,374,000.00 *gal*
Application event totals		144.85	12.69	141.43	2,200.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	42.87	3,888,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	40.19	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	40.19	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.19	

Field 1 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	37.51	3,402,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	34.84	3,159,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.84	

Field 2 - 11/29/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.24	25.57	372.11	1,713.76	2,250,000.00 *gal*
Application event totals		123.24	25.57	372.11	1,713.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	51.84	9,234,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.54	11.29	173.19	1,682.12	1,425,000.00 *gal*
Application event totals		60.54	11.29	173.19	1,682.12	

Field 2 - 06/27/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.10	13.07	200.53	1,947.72	1,650,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	60.02	10,692,000.00 *gal*
Application event totals		70.10	13.07	200.53	2,007.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	51.84	9,234,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	155.67	13.64	152.00	2,313.46	1,425,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	51.84	9,234,000.00 *gal*
Application event totals		155.67	13.64	152.00	2,365.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	51.84	9,234,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.84	

Field 2 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	49.11	8,748,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.70	8,140,500.00 *gal*
Application event totals		0.00	0.00	0.00	45.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.65	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.65	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.65	

Field 3 - 12/28/2016: Almond, in shell

Field name: Field 3

Crop: Almond, in shell

Plant date: 12/28/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	29.58	5,748,189.00 *gal*
Application event totals		0.00	0.00	0.00	29.58	

Field 3 - 12/28/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	190.09	32.38	448.76	3,159.52	300.00 *ton*
Application event totals		190.09	32.38	448.76	3,159.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	59.16	11,496,378.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	26.88	0.00	0.00	0.00	
Application event totals		26.88	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	59.16	11,496,378.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	26.88	0.00	0.00	0.00	
Application event totals		26.88	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	59.16	11,496,378.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Field 3 - 12/28/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	59.16	11,496,378.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	59.16	11,496,378.00 *gal*
Application event totals		0.00	0.00	0.00	59.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	29.58	5,748,189.00 *gal*
Application event totals		0.00	0.00	0.00	29.58	

Field 4 - 12/19/2017: Almond, in shell

Field name: Field 4

Crop: Almond, in shell

Plant date: 12/19/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	27.27	4,946,326.00 *gal*
Application event totals		0.00	0.00	0.00	27.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	190.09	32.38	448.76	3,159.52	280.00 *ton*
Application event totals		190.09	32.38	448.76	3,159.52	

Field 4 - 12/19/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	54.54	9,892,652.00 *gal*
Application event totals		0.00	0.00	0.00	54.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	26.88	0.00	0.00	0.00	
Application event totals		26.88	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	54.54	9,892,652.00 *gal*
Application event totals		0.00	0.00	0.00	54.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	26.88	0.00	0.00	0.00	
Application event totals		26.88	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	54.54	9,892,652.00 *gal*
Application event totals		0.00	0.00	0.00	54.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	54.54	9,892,652.00 *gal*
Application event totals		0.00	0.00	0.00	54.54	

Field 4 - 12/19/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	54.54	9,892,652.00 *gal*
Application event totals		0.00	0.00	0.00	54.54	

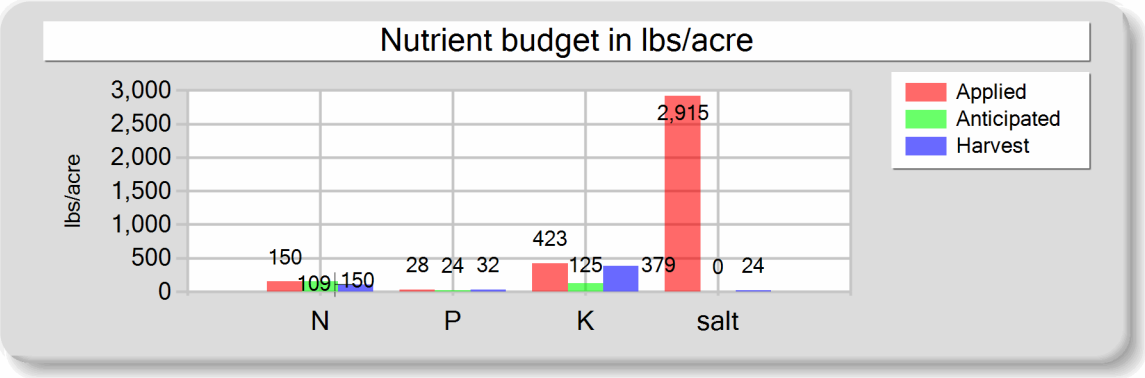
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	27.27	4,946,326.00 *gal*
Application event totals		0.00	0.00	0.00	27.27	

B. NUTRIENT BUDGET

Field 1 - 11/29/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	143.28	28.41	422.69	2,861.14
Fresh water	0.00	0.00	0.00	53.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.28	28.41	422.69	2,914.74
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	108.56	31.94	379.00	23.61
Nutrient balance	41.72	-3.53	43.69	2,891.13
Applied to removed ratio	1.38	0.89	1.12	123.47

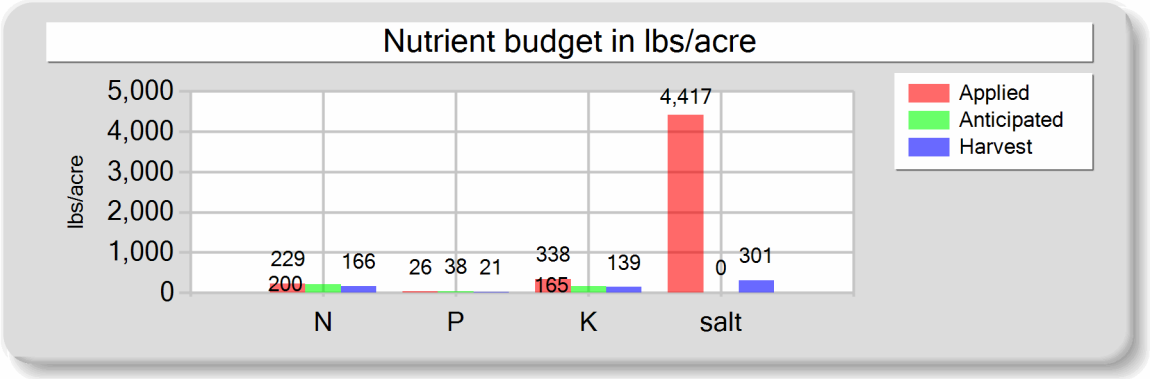
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
6.39	*inches/acre*

Process wastewater applied	
1,500,000.00	*gallons*
55.24	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/27/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.69	25.53	338.37	4,065.51
Fresh water	0.00	0.00	0.00	351.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.69	25.53	338.37	4,416.54
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	166.00	21.34	139.44	300.79
Nutrient balance	62.69	4.19	198.93	4,115.75
Applied to removed ratio	1.38	1.20	2.43	14.68

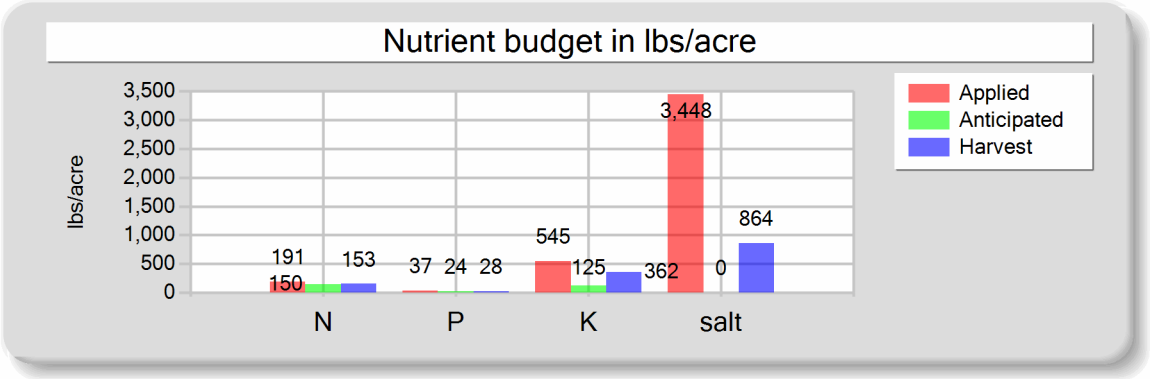
Fresh water applied	
31,833,000.00	*gallons*
1,172.30	*acre-inches*
41.87	*inches/acre*

Process wastewater applied	
1,500,000.00	*gallons*
55.24	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/29/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.78	36.86	545.30	3,395.88
Fresh water	0.00	0.00	0.00	51.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.78	36.86	545.30	3,447.72
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	153.03	27.64	361.83	864.33
Nutrient balance	37.75	9.22	183.46	2,583.39
Applied to removed ratio	1.25	1.33	1.51	3.99

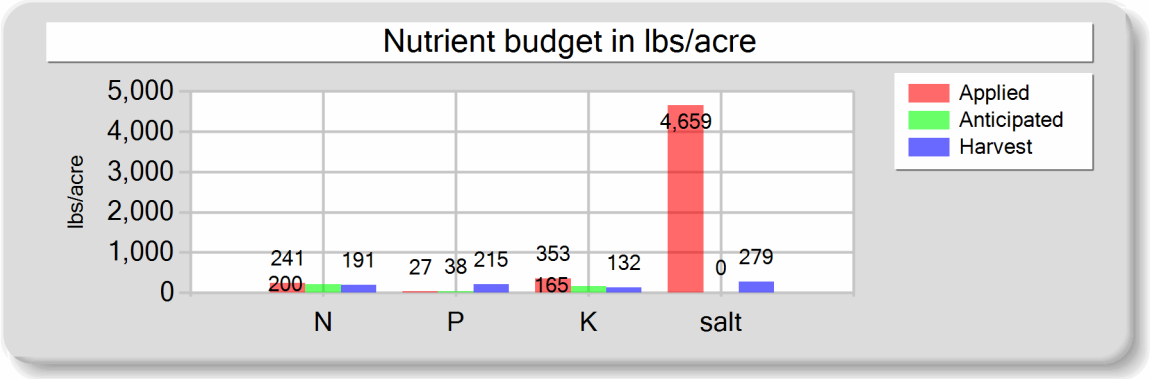
Fresh water applied	
9,234,000.00	*gallons*
340.06	*acre-inches*
6.18	*inches/acre*

Process wastewater applied	
3,675,000.00	*gallons*
135.34	*acre-inches*
2.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/27/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.77	26.71	352.53	4,261.18
Fresh water	0.00	0.00	0.00	397.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.77	26.71	352.53	4,658.84
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	190.89	215.36	132.15	278.99
Nutrient balance	49.88	-188.65	220.37	4,379.85
Applied to removed ratio	1.26	0.12	2.67	16.70

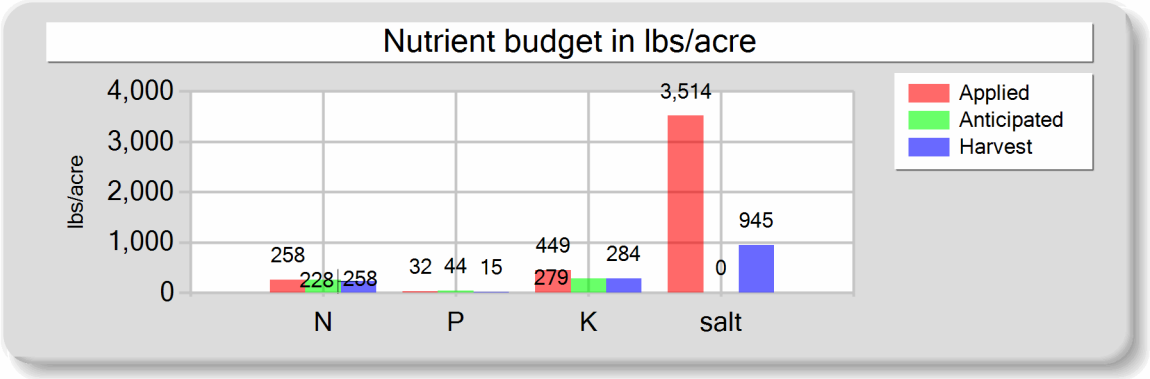
Fresh water applied	
70,834,500.00	*gallons*
2,608.59	*acre-inches*
47.43	*inches/acre*

Process wastewater applied	
3,075,000.00	*gallons*
113.24	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 12/28/2016: Almond, in shell

Field name: Field 3 Crop: Almond, in shell Plant date: 12/28/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	53.76	0.00	0.00	0.00
Dry manure	190.09	32.38	448.76	3,159.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	354.97
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	257.85	32.38	448.76	3,514.49
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	228.47	15.00	284.17	945.41
Nutrient balance	29.38	17.37	164.59	2,569.08
Applied to removed ratio	1.13	2.16	1.58	3.72

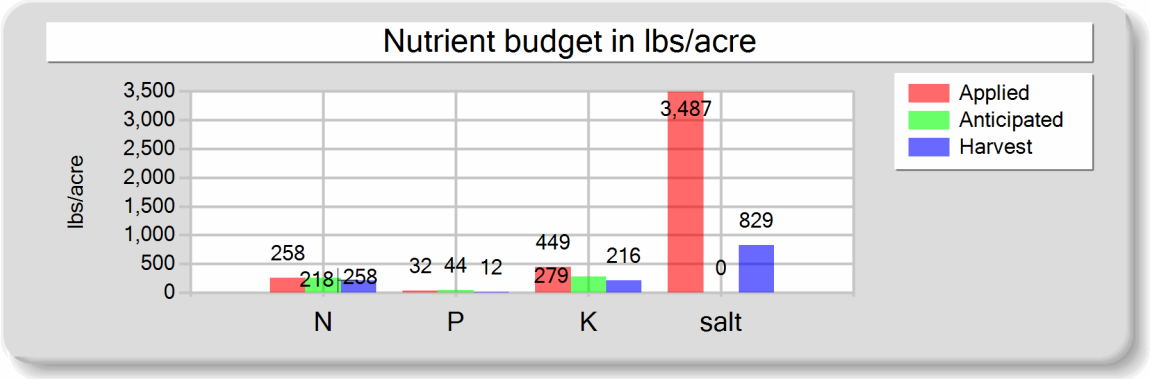
Fresh water applied	
68,978,268.00	*gallons*
2,540.24	*acre-inches*
42.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 12/19/2017: Almond, in shell

Field name: Field 4 Crop: Almond, in shell Plant date: 12/19/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	53.76	0.00	0.00	0.00
Dry manure	190.09	32.38	448.76	3,159.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	327.27
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	257.85	32.38	448.76	3,486.79
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	218.07	12.48	216.04	828.69
Nutrient balance	39.78	19.90	232.71	2,658.10
Applied to removed ratio	1.18	2.60	2.08	4.21

Fresh water applied	
59,355,912.00	*gallons*
2,185.88	*acre-inches*
39.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	3,730.00	51,700.00	0.00	0.00	0.00	0.00	0.00		36.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,000.00	11,300.00	45,900.00	0.00	0.00	0.00	0.00	0.00		39.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	4,920.00	34,500.00	0.00	0.00	0.00	0.00	0.00		43.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	361.00	132.00	0.00	0.00	74.90	1,090.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,580.00	5,020
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,500.00	308.00	0.00	0.00	177.00	1,280.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,910.00	3,940
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	280.00	157.00	0.00	0.00	52.20	801.00	120.00	115.00	229.00	2,200.00	0.00	248.00	0.10	5,120.00	7,780
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	720.00	146.00	0.00	0.00	63.10	703.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,090.00	10,700
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	462.00	120.00	0.00	0.00	75.10	1,030.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,940.00	4,520
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

MID Lateral 3

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.70	37
DL	1.00	1.00	1.00	0.50	0.50	0.50	0.50	2.00	1.00	0.10	10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/29/2022: Oats, silage-soft dough

Field 1 - 11/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,010.00	1,180.00	14,000.00		0.20
DL	100.00	100.00	100.00		0.01

Field 1 - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	450.00	2,940.00		2.10
DL	100.00	100.00	100.00		0.01

Field 2 - 11/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,090.00	1,100.00	14,400.00		9.10
DL	100.00	100.00	100.00		0.01

Field 2 - 11/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,570.00	1,200.00	17,600.00		8.18
DL	100.00	100.00	100.00		0.01

Field 2 - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	4,400.00	2,700.00		1.90
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	500.00	3,200.00		1.90
DL	100.00	100.00	100.00		0.01

Field 3 - 12/28/2016: Almond, in shell

Field 3 - 12/28/2016: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	1,320.00	25,000.00		9.10
DL	100.00	100.00	100.00		0.01

Field 4 - 12/19/2017: Almond, in shell

Almond

Sample and source description: Almond

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	1,230.00	21,300.00		8.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

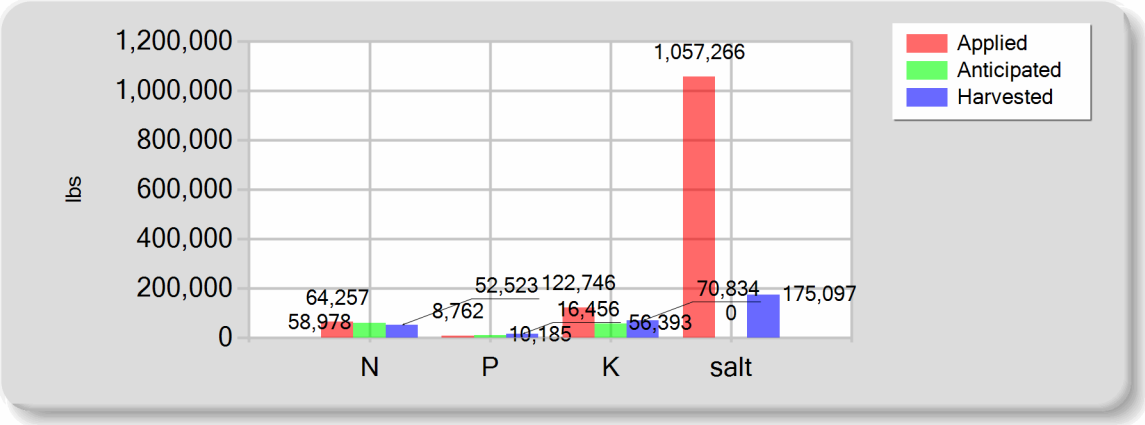
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

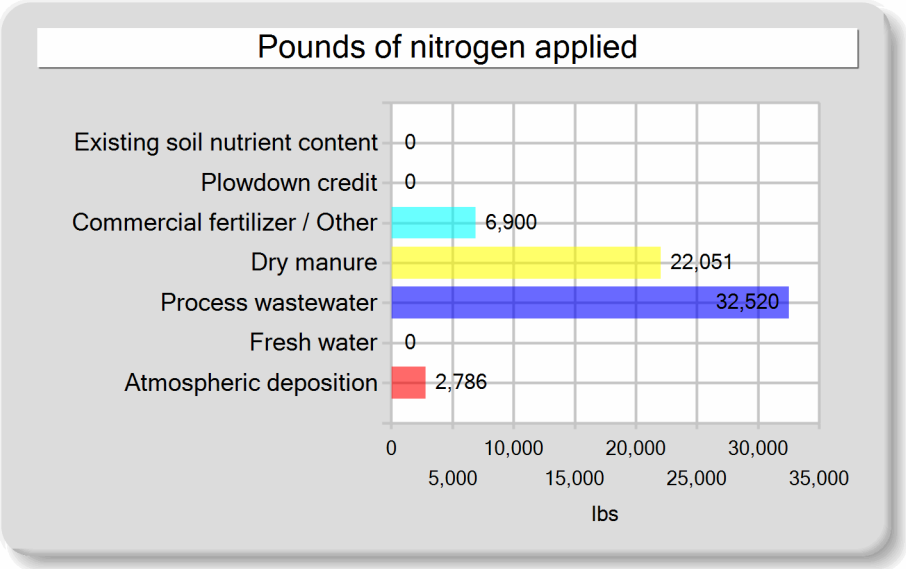
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

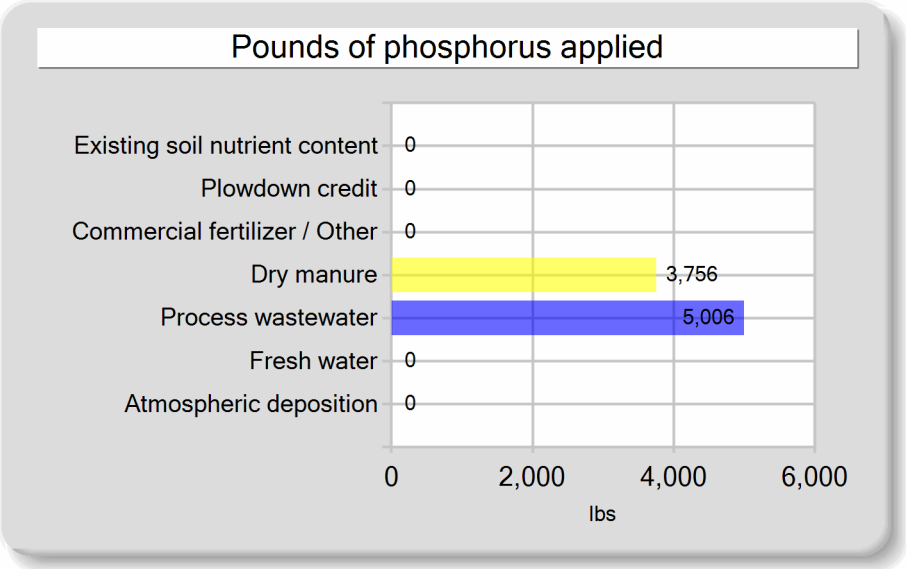
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	6,900.16	0.00	0.00	0.00
Dry manure	22,050.67	3,755.66	52,055.70	366,504.32
Process wastewater	32,520.47	5,006.50	70,690.08	615,084.92
Fresh water	0.00	0.00	0.00	75,676.97
Atmospheric deposition	2,786.00	0.00	0.00	0.00
Total nutrients applied	64,257.30	8,762.16	122,745.77	1,057,266.20
Anticipated crop nutrient removal	58,978.00	10,185.30	56,392.50	0.00
Actual crop nutrient removal	52,522.96	16,455.92	70,833.92	175,096.99
Nutrient balance	11,734.34	-7,693.76	51,911.85	882,169.21
Applied to removed ratio	1.22	0.53	1.73	6.04

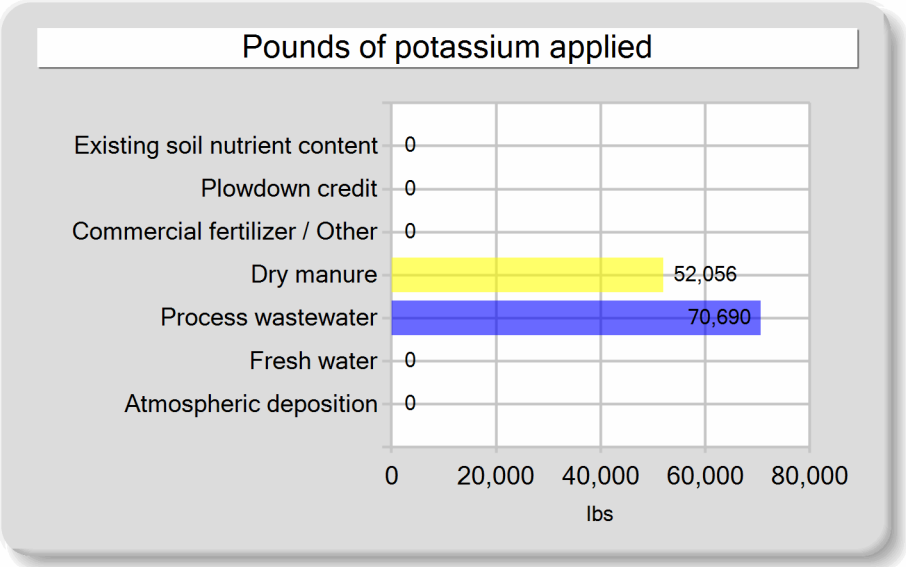
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

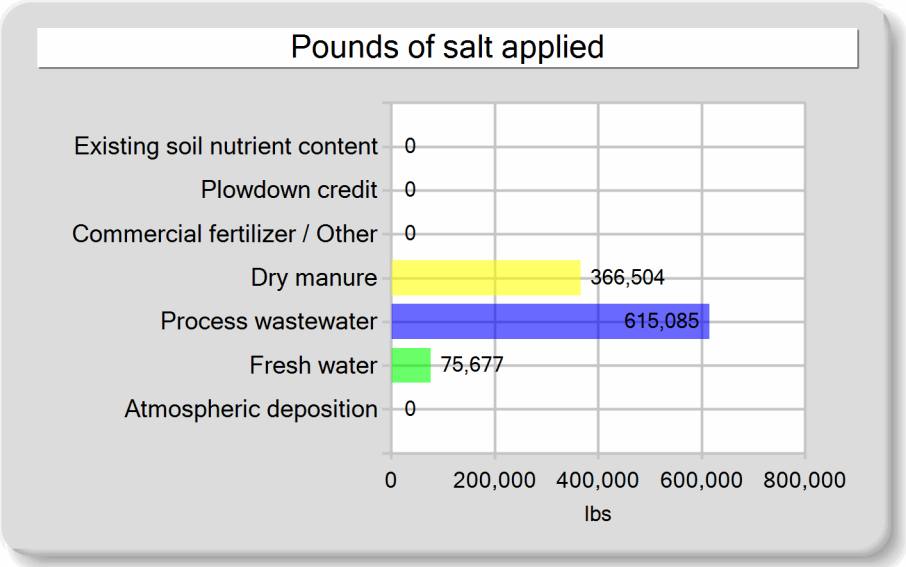


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

All animals were removed from the facility in October 2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe Henriques	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-6-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: JOE HENRIQUES
Mailing address:
(not facility address) MODESTO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	3/23/23	10:50	DW - MILK BARN	DOMESTIC	18.8		X				
2		10:55	DW - OLD WELL		19.0		X				
3		11:10	DW - GATES RD.		19.0		X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-23-23 3:50			T3082016

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Joe Henriques Dairy
556 N. Gates Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Milk Barn	T3082016A	Electrical Conductivity (EC)	EPA 120.1 and	0.942	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.00	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	603	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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www.denelelabs.com

Compliance Analysis Report

Customer
Joe Henriques Dairy
556 N. Gates Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Old Well	T3082016B	Electrical Conductivity (EC)	EPA 120.1 and	1.92	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1230	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	101			0.470
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

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Compliance Analysis Report

Customer
Joe Henriques Dairy
556 N. Gates Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Gates Rd.	T3082016C	Electrical Conductivity (EC)	EPA 120.1 and	0.942	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.885	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	603	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	101			0.470
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

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